

Computational Three Distances: Exploring the Aesthetics of the Southern Song Dynasty and Its Adapted Simulation and Rendering

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ABSTRACT

This paper references the painting theory of the "three distances" concept from the Southern Song Dynasty. It introduces a contemporary reinterpretation termed "Computational Three Distances," which employs computer algorithms as the means of painting, departing from traditional ink-brush techniques. The three distances encompass "micro-distance, volume-distance," and "time-distance." This study reexamines the adjustability of microscopic world units, the density of medium within volumes along with light-scattering effects, and the accumulation and erosion of time. It then puts this theory into practice, offering illustrative methods for each aspect.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts; • **Computing methodologies** → Modeling and simulation; Simulation types and techniques; Real-time simulation; Computer graphics; Rendering; Ray tracing; Computer graphics; Animation; Procedural animation.

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1 INTRODUCTION

The theory of three distances was originally proposed by Guo Xi (ca.1000-1087) as a compositional method for landscape painting, which includes "high distance, flat distance, and deep distance" [Guo, 2013]. This theory defines a multi-perspective approach to composition. Based on Guo's three distances theory, Han Zhuo (ca.1095-1123) proposed a new set of three distances of waterscape: "broad distance, hidden distance, and obscure distance" [Yu, 1957]. Han's three distances describes "broad distance" as the view of distant mountains from the near bank of a wide body of water,

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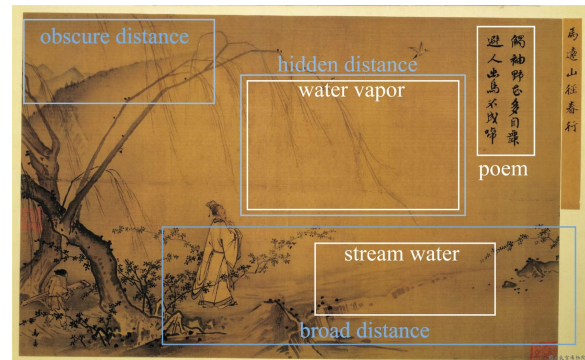


Figure 1: Explanation of Han's three distances with Ma Yuan's *Walking on a Mountain Path in Spring*, annotated by authors

"hidden distance" as the obscured view beyond a wilderness stream covered in mist and clouds, and "obscure distance" as a scene with fragmentary objects and an overall indistinct and shrouded appearance. Taking Ma Yuan's (ca.1190-1224) painting [Ma, ca. 1190-1224] as an example (see Fig. 1), in the composition, a small stream divides the two banks as "broad distance," while most of the scene is obscured by water vapor, creating a "hidden distance." The faintly visible distant mountains represent the "obscure distance." Chen Bao-zhen believes that Han's three distances is "a set of spatial representation methods created to present poetic scenery" [Chen, 1996]. Han explores ten different forms of water vapor and provides specific descriptions. Han mentions that from the intangible wind to the effects of smoke and clouds caused by different concentrations of media, a poetic atmosphere is created [Li, 2019]. We can see that Han associates the natural changes in weather with poetic sentiment, believing that water vapor can create poetry in artworks.

Ideologically, the Song dynasty can be considered a period that highly esteems the beauty of "plainness." "Plainness" implies simplicity and moderation, representing the shared preference for simplicity and neutrality in both Taoism and Confucianism. By examining landscape paintings from the Southern Song dynasty, most notably exemplified by the compositional techniques of Ma Yuan's "one-corner" and Xia Gui's (ca. 1180-1224) "half-side," we can observe that water vapor is frequently employed to establish voids and minimize visual details, thereby presenting a style that aligns with the concept of "plainness." Moreover, these blank spaces are perceived as embodying boundless and abundant qualities.

This paper argues that Han's three distances is not only applicable to ink-washing paintings but, to the development of modern CG and programming languages. Han's theory inspires us to reexamine material characteristics from different perspectives. Through the lens of "broad distance," we perceive that the expansiveness achievable in digital media lies in the transformation of units, allowing us to observe different levels of imagery through distinct units. The "hidden distance" approach, achieved through the concealment of water vapor, resonates with volume rendering techniques, prompting us to focus on the relationship between the medium filling the space and light. The concept of "obscure distance," describing distant objects, draws attention to the interplay between time and imagery. We believe that time is a prominent feature of digital media, enabling us to reconsider the presentation of imagery from a temporal perspective.

2 RELATED CONCEPTS AND WORKS

This paper chose the association between water and atmosphere as the main focus and explored Han's three distances through the works of two artists: Ma Yuan's *Water Studies* [Ma, ca. 1160-1225] and Hiroshi Sugimoto's *Seascapes* series of photographs [Sugimoto, 2015]. Ma was one of the few painters of the Southern Song period who focused on water as the main theme in his works. He employed traditional ink-washing painting techniques such as the "cun" technique (texture wrinkles) and the use of blank spaces to depict twelve different forms of water. By using the one-corner technique, he ensured a consistent perspective in his paintings and removed unnecessary details, allowing viewers to concentrate on the myriad expressions of water. When people see his works, they are often captivated by the variations in brushwork and hope to recreate the artist's unique style in the field of CG. However, what we are more interested in is the composition of his works. Ma utilized a significant amount of blank space to create a poetic and concise appearance, with the infinite extension of this blank space representing the concept of "hidden distance" in Han's three distances. Another representative work is Sugimoto's *Seascapes* series created between 1980 and 2002. These works deliberately exclude objects other than the ocean and the atmosphere, resulting in simple yet poetic images. Sugimoto asserts that seascapes hold significance as the sole landscapes connecting us to the ancients in our modern world. We believe that by observing Sugimoto's works through Han's three distances, we not only encounter the shared theme of water and atmosphere but also witness visually clean and poetic compositions. Furthermore, Sugimoto further explores the concepts of time and memory, which inspires us to incorporate time and memory into our proposed Computational Three Distances (CTD) theory.

Regarding the study of ink-washing media or Shanshui painting themes in the field of digital media, we have found that most of the technical research and artistic works primarily focus on the development of non-photorealistic rendering styles, aiming to replicate the appearance of ink-washing paintings, such as brushstrokes. In terms of non-ink-washing visual styles, there are studies on multi-viewpoint compositions for presenting ink-washing paintings [Li, 2011] and Ukiyo-e prints [Yoshimura and Saito, 2017] in

3D space. The emphasis of this study, however, lies not only in exploring the aesthetic and humanistic reflections within these works but also dives into the corresponding algorithmic techniques of microstructures, volumetric media, and time in CG technology.

3 COMPUTATIONAL THREE DISTANCES

Following Han's three distances, we propose CTD: "*micro-distance*," "*volume-distance*" and "*time-distance*". The theoretical and aesthetic elaboration of CTD is presented in Chapter 3, while the practical techniques are discussed in Chapter 4.

3.1 Micro-distance

Corresponding to "broad distance," it originates from the microscopic perspective of microfacet models [Walter et al., 2007]. Based on raymarching rendering, pixels on the screen are computed as rays projected onto microfacets, combining their secondary reflections and refractions. This approach allows us to exhibit the appearance and subtle variations of subsurface materials, which we perceive as the breadth in depth within the microscopic world. Furthermore, we can regard reflections and refractions as a form of microscopic units. The units of the microcosmic world can also be controlled by the creators, selecting different scales according to the content of their creation to emphasize the intended meanings. Thus, we are able to transcend the limitations of physical distance and "witness a broader world," similar to observation through a microscope. This includes freely manipulating viewpoints in 3D space to appreciate the sculptural observation of the microcosmic structures.

3.2 Volume-distance

Volume-distance is related to "hidden distance" and is a technique used to create fog and translucency effects. By employing atmospheric scattering [Nishita et al., 1996] and volumetric rendering [Schneider, 2017], the density varies based on the distance. Scattering of light and blending of colors can generate a blurred visual effect, as if the viewer is observing through a thin mist. In literati Shanshui paintings, the "hidden distance" effect is achieved by reducing depiction, rendering, or using sparse brushwork. The space created by this reduction in depiction is actually filled with a medium, making the medium the primary subject of depiction in volume-distance. Different mediums can be designed to fill the space, and the interaction between the mediums and light can produce various visual effects.

3.3 Time-distance

The concept of "obscure distance" refers to a distance so far that it becomes almost indiscernible. In contrast to the tangible existence and richness of volume-distance, the concept of obscure distance creates a blank space on the canvas with minimal information. We extend the notion of distance from space to time, transforming the distance in space into the duration in time, which we refer to as time-distance. Dynamic representation in digital media can unveil different layers of temporal significance, and the concept of time can be presented through methods such as real-time rendering, long-term exposure, and sampling over time intervals to capture the unique appearance of a subject at specific moments in time. In

the concept of time-distance, we aim to emphasize the accumulation and erosion effects of time, where the imagery can depict the distinct dynamic texture and appearance generated by accumulations and intervals of the material world over time.

4 IMPLEMENTATION AND DISCUSSION

CTD was refined through *Imitation of Shanshui*, a series of generative artworks that simulate dynamic and interactive 3D scenery such as sea, air, and land. This simulation is based on OpenGL shading language, Signed distance function, and Raymarching rendering. The *Imitation of Shanshui* series includes artworks like *The Chosen Paysage* (2022), *The Odd Paysage* (2020), and *The Ancient Paysage* (2018). This project originated from a collaborative new media art exhibition with the National Palace Museum, utilizing procedural simulations and rendering methods as a novel medium for landscape painting in the digital era. *The Ancient Paysage* depicts a seascape using ocean simulation and subsurface scattering. *The Odd Paysage* portrays cloudscape using volumetric rendering with fractional Brownian motion (fBm) noise-based density. *The Chosen Paysage* presents the landscape using fBm procedural terrain and hydraulic erosion.

In Ch.4, we specifically select significant scenes from *The Ancient Paysage* – namely, *Blue Tears* and *Eternal* – to explicitly demonstrate the techniques of CTD. Furthermore, throughout the *Imitation of Shanshui* series, CTD is applied, enhanced, and diversified through various methods.

4.1 Micro-distance

In 3D space, the depth of space is composed of refractions, reflections, scattering, and dispersion of light. For example, the gradual shading of waves, reflections, and refractions on wave crests, and the clarity on surfaces facing or backlit by light. We utilize Fresnel reflection based on microfacet-BRDF to emphasize reflection or undersurface refraction.

$$C_{Final} = F * C_{Reflected} + (1 - F) * C_{Refracted}$$

$$F = f_0 + (1.0 - f_0) * pow(1.0 - dot(H, V), 5.0)$$

where C stands for color, F stands for Schlick's Fresnel approximation, V stands for view direction, H stands for half vector of V and light direction, f_0 is reflectance at normal incident. In *Blue Tears* (see Fig. 2), we utilize N, the surface normal, instead of H mentioned in the above equation, as we focus on the macro-surface related to the viewing angle. The precise calculation of shining reflection with implicit subsurface luminescence is of great importance. The transparency conveyed through optical refraction is not easily depicted in traditional media, but it can be accurately represented through digital media and computing means.

4.2 Volume-distance

Building on the previous representation of the relationship between objects and the atmospheric ambiance using volume-distance, the weighting of the medium can be adjusted based on the theme and desired visual effects. Therefore, Density(p) becomes a focal point in volume-distance. Building upon the foundation of raymarching, we further employ volume rendering to depict the volume and thickness of water vapor. Due to the presence of a medium filling

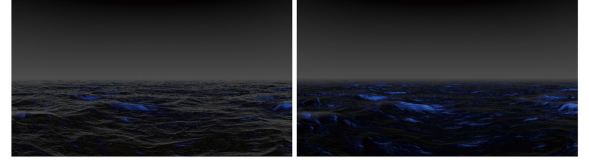


Figure 2: Higher Fresnel approximation stresses on reflection(left); lower Fresnel approximation stresses on refraction(right). (*Blue Tears* ©Chi-min Hsieh)

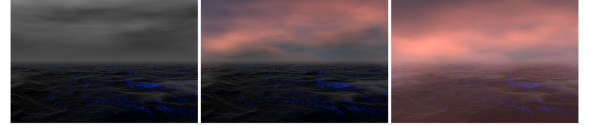


Figure 3: In comparison with traditional texture-based clouds in high-level(left), applying non-uniform density enhances volumetric clouds in the mid-level(middle) or vivid mist on surface-level(right). (*Blue Tears* ©Chi-min Hsieh)

the space, the amount of information and computational complexity involved exceeds the previous stages of volume rendering. According to Beer's Law, the attenuation of light is directly proportional to the distance and the density of the medium, and it is calculated through volume rendering as follows:

$$T_{transmittance}(p) = \exp(-d_{distance}(p) * coef. * Density(p))$$

$$I_{illumination}(p) = T_{transmittance}(p) * L_{light}(p) * s$$

where T stands for transmittance, p stands for the position of each step d stands for the distance between camera and p, coef. stands for extinction coefficient, I stands for illumination of volume rendering, L stands for light intensity, and s stands for step size. Density(p) can be uniform constant or Density(p) = weightCoef * SDF(p). SDF(p) stands for a model defined by signed distance function. Another distinctive feature is the design of a non-uniform and spatially dependent density function. Taking inspiration from Inigo Quilez's warping noise[Quilez, 2019], we design the following formula: space-varying Density(p) = fBm(p + fBm(p + fBm(p))) * weightCoef * SDF(p). As shown in Figure 3, designing complex and non-uniform density of water vapor can express subtle nuances of clouds and explore the interplay of volume and atmosphere.

4.3 Time-distance

We drew inspiration from time-lapse and time-averaging techniques in photography. Borrowing the concept of accumulation from long exposures and incorporating the idea of uneven erosion over time, we captured the essence of time passing. Through stacking and sculpting frames, we were able to capture the flow of time. As seconds and minutes ticked away, image details gradually merged, boundaries between elements faded, and prominent differences were neutralized. Dynamic accumulation is achieved using a recursive accumulation framebuffer, represented as follows: where I stands for image color, α stands for mixing weight.

$$I_{framebuffer}(t) = \epsilon(p, t) I_{image}(t) + (1 - \alpha(p, t)) * I_{framebuffer}(t - 1)$$

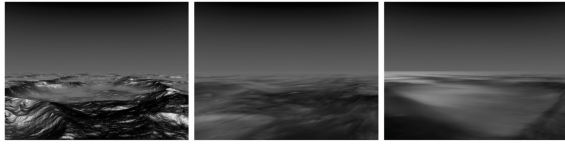


Figure 4: The left image($p, t=0.8$) is an instantaneous movement of the sea; the middle image is a short exposure transforming the ocean into a misty look; the right image $\alpha(p, t)=0.005$ is a sea of clouds created with long exposure. (*Eternal* ©Chi-min Hsieh)

If $\alpha(p, t)$ is a fixed value, for example, a value of 0.1 would produce a blurry visual effect, while a value of 0.8 would maintain a clear appearance. The *Eternal* scene (see Fig. 4) showcases the impact of time on visual texture, transforming the ocean in the image into a seascape of clouds as time progresses. The unique design of this work involves setting $\alpha(p, t)$ to very small values, ranging between 1/1000 and 5/1000. This leads to the slow but continuous accumulation of dynamic textures. The visual texture of the ocean transitions into a sea of clouds. Furthermore, by further manipulating the dynamic values, more complex visual representations can be generated.

4.4 Discussion

Micro-distance can be regarded as a technique for adjusting microscopic units, affecting the range of the field of view. In Section 4.1, we presented examples of surface and subsurface effects. Here, we would like to provide another concept (see Fig. 5), where we consider dynamics as units, emphasizing the exaggerated representation of water vapor with significant movement. Building upon the micro-distance, we further employ volume rendering to depict the volume and thickness of water vapor, which involves a greater amount of information and computational complexity compared to the previous stage of volume rendering. Volume and time-distance can be seen as methods to achieve the visual effect of "plainness," causing the image to not only blur over time but also occasionally condense and dissipate, exhibiting the characteristic of material transformation distorted by time. In Han's three distances theory, he employs water vapor and distance to convey "plainness" and poetic sentiment. The specific approach involves leaving blank spaces on the canvas, creating a clean visual effect. On the contrary, our works convey the philosophical meaning of "plainness" through the layering of visible and invisible mediums, spaces, and time (see Fig. 6). For us, the distance, accumulation, and erosion of time itself are elements of material transformation and possess poetic qualities. Therefore, we believe that CTD successfully conveys the visual characteristics and philosophic meaning of "plainness" through CG technology.

5 CONCLUSION

CTD presents another digital representation of Southern Song Dynasty Shanshui paintings, not by replicating the texture or brushstrokes of ink-washing paintings, but by expanding Han's three distances to showcase the aesthetic core of "plainness." CTD allows

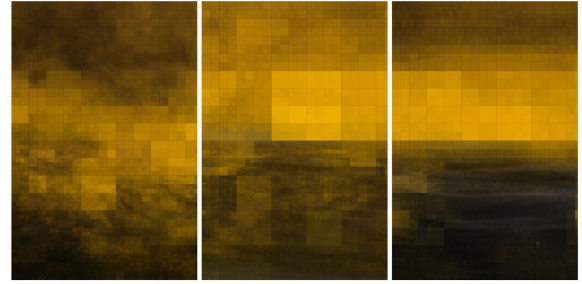


Figure 5: A variation of micro-distance to exaggerate invisible vapor. (*The Odd Paysage* ©Chi-min Hsieh)

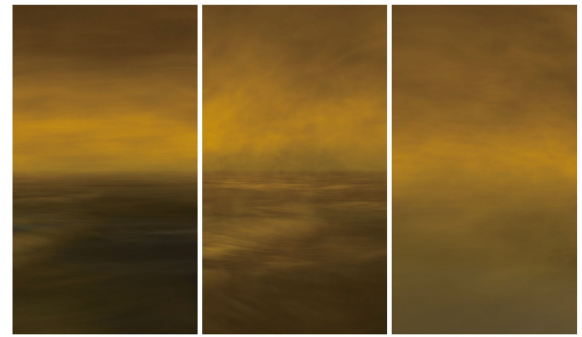


Figure 6: Achieving "plainness" without using blank space, by filling the space through the stacking of mediums and time. (*The Odd Paysage* ©Chi-min Hsieh)

us to rethink existing digital media techniques from different perspectives and provide new creative inspirations. The viewpoint of micro-distance enables us to break free from the limitations of a single perspective ratio and use macro and micro levels more flexibly, even blending and intertwining multiple layers. The viewpoint of volume-distance explores different coefficients, densities, and distances of medium particles to create the visual effect of "plainness" and imbue it with rich substantial connotations. The concept of time-distance allows us to approach the artwork from a temporal perspective, simulating the accumulation or erosion of the medium over time and even echoing Shimamoto's discourse on depicting collective memory through temporal accumulation.

While traditional ink-washing achieves the visual effect of "plainness" through the reduction in depiction, in the field of CG, we can utilize the stacking and subtraction techniques of volume-distance and time-distance. "Plainness" that creates simple and elegant visual effects can not only be presented in 2D but also be extended into 3D space, showcasing depth and layers suitable for immersive experiences. We believe that applying CG techniques to express the beauty of Han's three distances can convey concepts of time and spatial dimensions, inspiring creators to perceive and express time and environmental space within the vast history and evolution of media.

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